Student dropout warnings.

How accurate are they,
and who gets missed?

University of Edinburgh

Predictive signal. Unequal errors. A question of support.

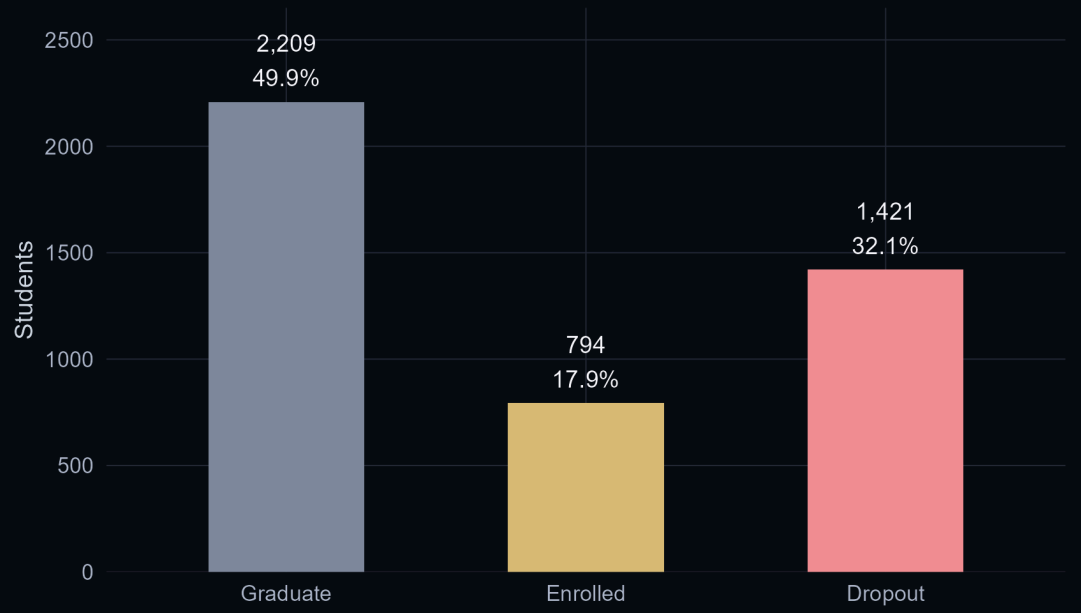
UCI 697 / ONE PORTUGUESE INSTITUTION
Analysis 05 NOV 2026 / Revision 11 NOV 2026

What counts as recorded dropout?

4,424

student records.
Three outcomes kept visible.

A course or institution change also counts as **Dropout**.

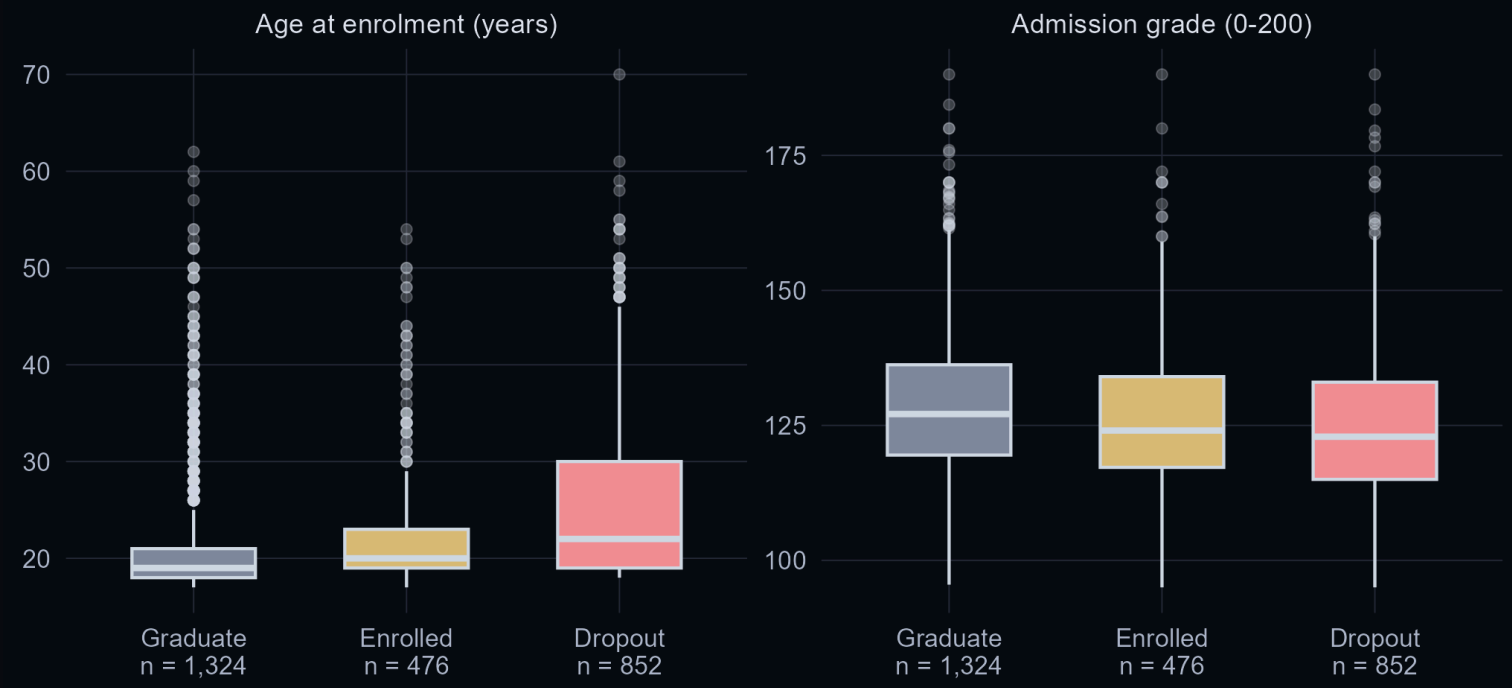


Target: recorded dropout by the normal course endpoint.

Graduate and Enrolled mean no recorded dropout then. Enrolled does not mean graduated.

Realinho et al. (2022), pp. 1-4. Enrolments 2008/09-2018/19; endpoint usually 3 years, 4 for Nursing. No causal or Edinburgh-wide claim.

Entry profiles overlap



Higher median age in the recorded-dropout group.
Substantial overlap in admission grades. **No individual outcome is determined.**

Training only, n = 2,652; associations, not causes.

Quiz 7 recommended Data to Viz; this prompted our figure review. Boxplot advice.

Training, validation and test

Age | Admission grade | Course | Logistic regression → probability

01 / FIT

2,652

Training

Fit the model.

02 / CHOOSE

883

Validation

Choose one warning threshold.

03 / EVALUATE

889

Test

Evaluate the frozen rule.

Course is categorical; age and grade are additive on the log-odds scale, without interactions.

Later and timing-uncertain inputs are excluded. Recorded gender is reserved for the audit.

> The regression lectures help us review these choices. Logistic regression required further learning.

Seed 20261105; approximately 60/20/20, stratified by original outcome and recorded gender. Model fitting and validation use no test outcomes.

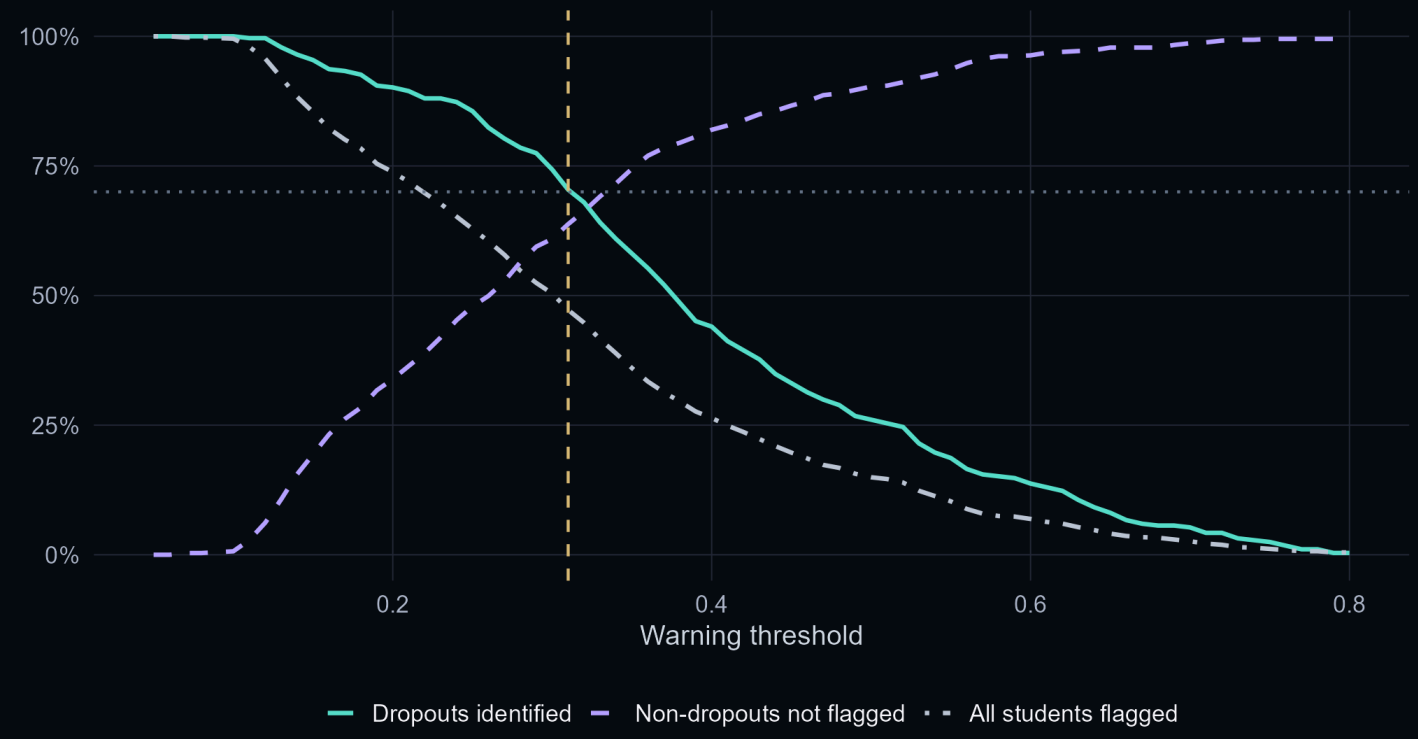
Choosing the warning threshold

0.31

One common threshold.

Highest specificity while identifying at least **70%** of validation dropouts.

Validation sensitivity:
200 / 284 = 70.4%



Our illustrative target, not a university policy. Grid 0.05-0.80 in steps of 0.01; one threshold for both groups. We then froze the rule and evaluated its sensitivity, precision and errors on test records.

Sensitivity and precision

65.6%
Sensitivity / recall
187 / 285 dropout records found

45.7%
Precision / PPV
187 / 409 warnings matched

64.0%
Accuracy
569 / 889 decisions matched

RULE	SENSITIVITY	PRECISION (PPV)	ACCURACY
Always no warning	0.0%	Not defined	67.9%
Threshold 0.50	28.1%	64.0%	71.9%
Validation rule: 0.31	65.6%	45.7%	64.0%

0.50 to 0.31: **+107** dropout records found; **+177** false warnings. **98** dropout records still missed.

Always no warning: **67.9% accuracy**, but no dropouts found. Who does our rule miss?

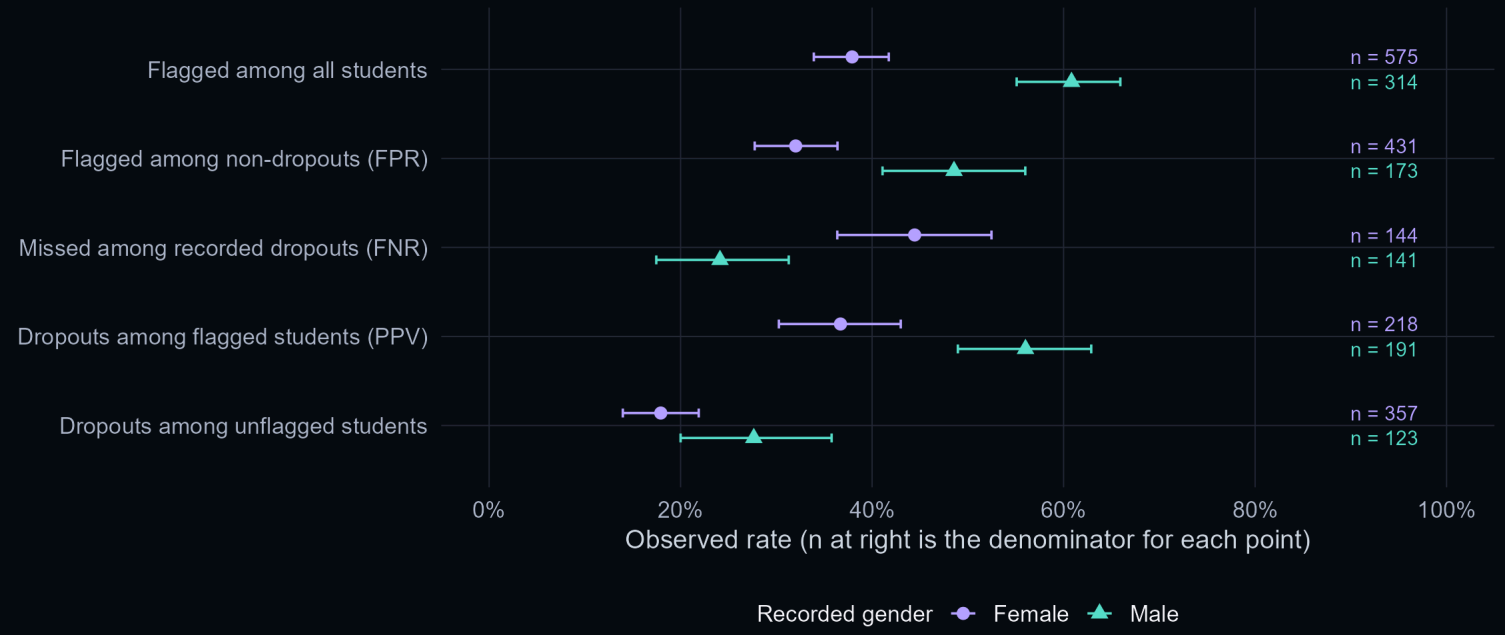
Test n = 889; 222 false warnings; 409 / 889 (46.0%) flagged. Sensitivity 95% conditional bootstrap interval: 60.3% to 71.0%. A false warning is relative to the recorded label.

Who is missed by the same rule?

44.4%
Female: 64 / 144
missed / dropout records

24.1%
Male: 34 / 141
missed / dropout records

Of the 98 missed records, 64 were female and 34 male. We compare within-group rates, rather than counts alone.



Coverage: Independence | Errors: Separation | Outcomes within warned / unwarned groups: Sufficiency

False-warning rates: Female 32.0%; Male 48.6%.

Recorded dropout proportions: Female 25.0%; Male 44.9%. We investigate the differences without treating them as proof of discrimination.

Points: observed rates. n: denominator for each point. Bars: pointwise 95% percentile bootstrap intervals, conditional on the fixed model and threshold; 5,000 resamples within each group. These are not standard deviations.

A warning should start a conversation.

Keep human review.

Offer support without punishment; allow records to be corrected.

Check capacity.

46.0% flagged is a substantial support commitment.

Investigate the errors.

Review both group error patterns and what the label measures.

Test later cohorts.

Validate at the institution where the rule would actually be used.

Predictive signal is not evidence that warnings prevent dropout.

One institution, historical cohorts, no cohort/year field for a chronological test. The model assumes a common additive relationship; random hold-out evaluation is not prospective validation.

Sources and assistance

- Realinho et al. **UCI dataset 697**, DOI 10.24432/C5MC89. CC BY 4.0. Retrieved 5 November 2026.
- Realinho et al. (2022). **Predicting Student Dropout and Academic Success**, *Data* 7, 146. 10.3390/data7110146.
- IDS course materials: **Algorithmic Bias**, pp. 6, 16-19; **Quantifying uncertainty**, pp. 21, 39, 46-53.
- IDS regression lectures: **Introducing linear models**, pp. 22-28, 48-54; **Model checking and multiple linear regression**, pp. 24-41, 57-63.
- R documentation: binomial logistic regression with glm. Fairness definitions: Fairness and Machine Learning.
- Figure review: Data to Viz, prompted by Quiz 7; page-specific advice is recorded in the notes.
- Review tools: Sciwrite (Barba); Anti-defensive Writing; Kassis et al. (2026), Scientific Agent Skills.

We applied our course methods to data handling, plots, bootstrap and the fairness audit.
For unfamiliar methods, we used ChatGPT and documentation, then rewrote and checked the main R analysis.
ChatGPT also supported initial code/text drafts, source checks and slide formatting. We take responsibility for the final work.

Backup: the three fairness questions

CRITERION	COMPARE	FEMALE	MALE
Independence	Warning rate	37.9%	60.8%
Separation	False-positive rate	32.0%	48.6%
Separation	False-negative rate	44.4%	24.1%
Sufficiency	Dropout among flagged (PPV)	36.7%	56.0%
Sufficiency	Dropout among unflagged (1-NPV)	17.9%	27.6%

Recorded dropout prevalence: Female **25.0%**, Male **44.9%**. These criteria ask different questions. Equality is not assumed, and a numerical gap is not a causal finding about discrimination.

Backup: counts and denominators

GROUP	STUDENTS	TP	FP	FN	TN
Female	575	80	138	64	293
Male	314	107	84	34	89
Total	889	187	222	98	382

TP: warned and recorded Dropout. FP: warned, no recorded Dropout.
FN: not warned, recorded Dropout. TN: not warned, no recorded Dropout.

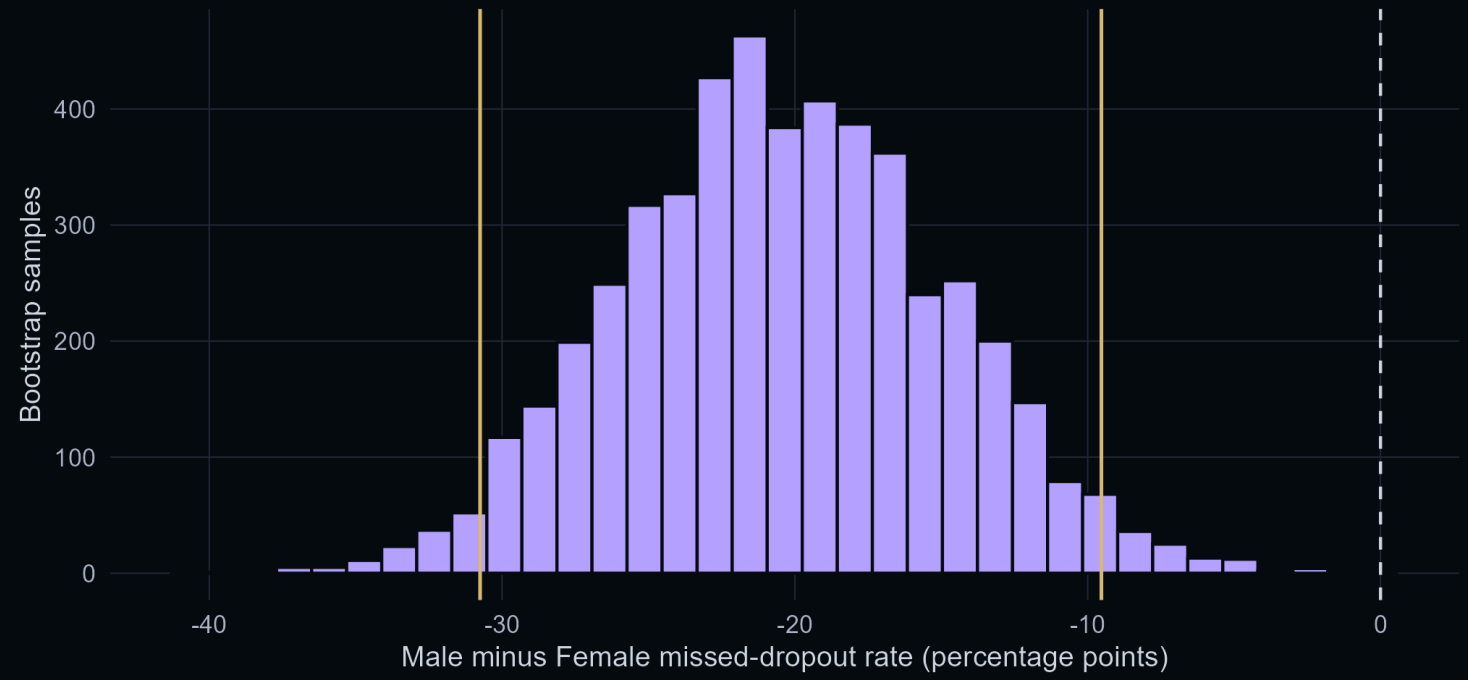
FPR = FP / (FP + TN). FNR = FN / (TP + FN).
PPV = TP / (TP + FP). NPV = TN / (TN + FN).

Backup: uncertainty in the group gaps



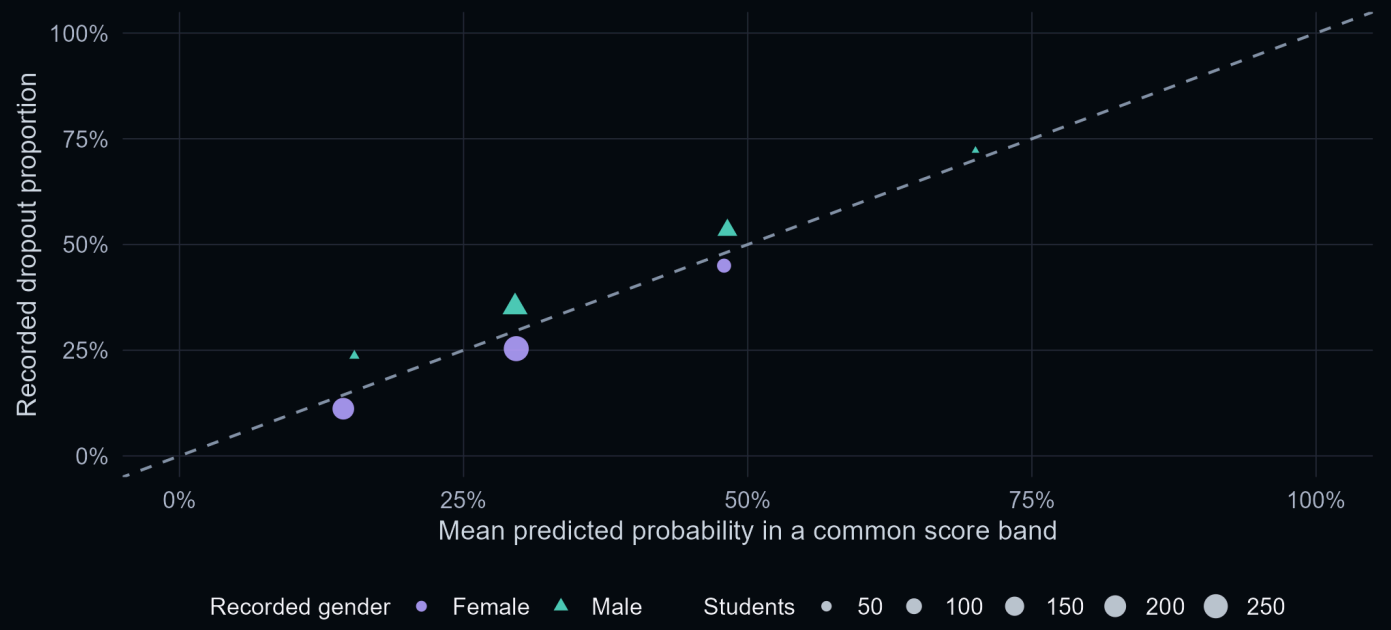
Resample held-out records within each recorded-gender group, recompute each metric, take the middle 95% of 5,000 gap estimates. No refitting, cohort-cluster correction or multiple-comparison adjustment.

Backup: the bootstrap we used



Gold lines are the 2.5th and 97.5th percentiles. Conditional on the fitted model and validation choice; the interval does not prove transportability or causal fairness.

Backup: checking predicted probabilities



We compare predicted probabilities with observed dropout proportions.

AUC checks ranking; Brier score checks probability error. Neither alone establishes calibration.

Colour and shape identify recorded gender; point size indicates n. Common 0.2-wide bands; n < 30 points omitted (retained in CSV). No intervals shown. This descriptive check establishes neither group calibration nor binary sufficiency.

Backup: sensitivity to the model inputs

INPUTSET	AUC	FEMALE FNR	MALE FNR
Primary: no gender	0.718	44.4%	24.1%
Add recorded gender	0.725	56.9%	9.2%

Both models use the **same training rows and threshold**.
The primary model was fixed before evaluating test performance.

Including gender changes the error pattern; it is not an automatic fairness correction. The slight AUC improvement does not decide which allocation of errors is acceptable.

Backup: why keep Enrolled?

ANALYSIS	TESTN	AUC	DROPOUTS IDENTIFIED
Primary: all three labels retained	889	0.718	65.6%
Primary fit on resolved test records	729	0.756	65.6%
Refit excluding Enrolled	729	0.758	81.8%

The last two rows evaluate only Dropout versus Graduate (729 test records) at the same numerical threshold. They change the population and question. Better-looking numbers cannot replace the all-student endpoint analysis.

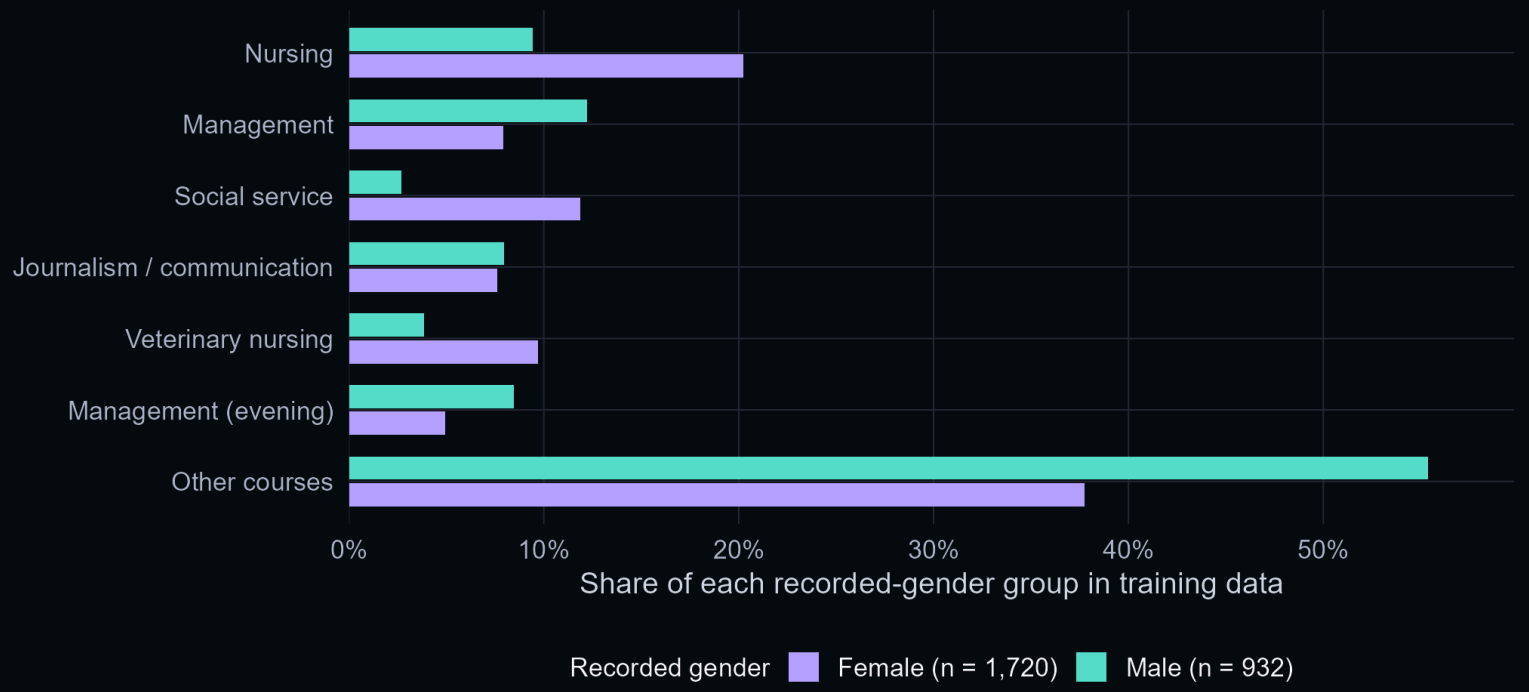
Backup: course connections and model assumptions

- **Course foundations:** data handling, factors, plots, bootstrap and the three fairness criteria.
- **Regression lectures:** multiple predictors, category contrasts, model checking and parsimony.
- **Fixed assumptions:** additive effects on the log-odds scale, without interactions. Interaction terms remain untested.
- **Further learning:** logistic regression, validation, thresholds, AUC and Brier score; ChatGPT and documentation.

We rewrote and ran the main R analysis and checked its outputs. We now use the newer regression lectures to review those earlier choices.

Introducing linear models, pp. 48-54; Model checking and multiple linear regression, pp. 24-41, 57-63. Details: docs/course-map and docs/regression-course-update.

Backup: course mix by group



Six largest courses, ordered by pooled training count; Other courses last (display only). Bars start at zero; each group's total is in the legend. This composition does not prove the cause of a test-set error gap or demonstrate Simpson's paradox.